

Lab 3: IMU Noise and Performance Characterization

Group 10: Savannah Macero & Siddhi Sanjay More

03/21/25

Discussion

1. The short stationary data shows that the VN-100 is subject to minimal noise that may be considered negligible, depending on its use case. Figure 1 in Appendix shows the time series of the gyroscope, accelerometer, magnetometer, and Euler angle data over approximately 5 minutes while placed on the ground in EXP. Note that the scale of the amplitude of the gyroscope data is much smaller than the three other signals, which is simply due to the nature of the signal types in order to represent all three axis values on the same plot. The gyroscope plot best visualizes the behavior for all four data types if you were to zoom in on a smaller time interval. 2. The mean, median, and standard deviation for roll pitch and yaw are reported below and in Figure 1:

	Roll	Pitch	Yaw
Median(deg)	-0.23	0.32	-46.05
Mean (deg)	-0.23	0.33	-46.04
Standard deviation (deg)	0.02	0.06	0.07

3. According to the VN-100 data sheet, the yaw, pitch, and roll readings can be estimated with a 1 Sigma uncertainty reported in degrees. This can be interpreted as a normal Gaussian distribution with approximately 68% of the measurements falling within one standard deviation of the mean. 4. While the data recorded falls slightly short of this metric, this could be expected to due location conditions at EXP. Vibrations, and magnetic disturbances from environmental factors like people walking by and the nearby Ruggles station are all possible sources of noise that affected the distribution of the dataset.

5. The video was started before and ended after the bag file was recorded so that the differences in the initial and final times of the video were recorded and could be identified by visual and audio cues in the video clip. Since the internal ROS2 clock was used for the bag recordings, the time array was normalized to $t_0 = 0$ by subtracting all time values by $t(0)$. To identify the time windows in the video containing the data of interest, the video timestamps were normalized by subtracting $t_{bag_start} = 13$ from all values, since $t(0)$ of the video is already 0. 6. The IMU data correlated well with the video of the actual motion. Linear motion in all three axes were performed by extended and retracting my arm from my chest, and then I rapidly circled the IMU about the Y axis. See Appendix B Figures 4 and 5 for plots.



Left to right: linear motion in Z axis, linear motion in X-axis, circular motion about the Y axis.

7. I performed Allan Variance analysis on Location D data. The plots of $\sigma(\tau)$, $\sigma_K(\tau)$, $\sigma_B(\tau)$, and $\sigma_N(\tau)$ are shown in Appendix C. The $\sigma(\tau)$ lines fit $\sigma_N(\tau)$ well in all axes of both the accelerometer and gyroscope data, suggesting that the dominant source of error during data collection was white noise. The accelerometer data in the Y and Z axes were initially dominated by white noise, but then becomes more influenced by rate random walk, as the slope becomes positive at the tail and fits well to $\sigma_K(\tau)$. This indicates that the data is mostly influenced by sources of noise with high frequencies and then begins to experience drift in Y and Z linear accelerations. The plots for Locations A, B, and C were received from Sofia Lanzi. Based off the four locations given, I would group A and B together to be the two locations with higher elevations, and C and D to be the two ground level/basement locations. The plots for A and B are much less linear and show influence from all three noise parameters, while C and D are primarily influenced by white noise and rate random walk. C experiences much more drift in both sensors and in all axes, opposed to D, which only experiences drift in the accelerometer. I would guess that C was recorded in ISEC, where sunlight can cause temperature changes and high traffic of people using electronic devices may increase random walk, opposed to the Snell library basement which may have been more static.

8. The results indicate that Allan variance analysis is a helpful tool to visualize how your sensor data is influenced by different types of noise, but to be able to interpret the results, it is important to know what noise to expect in the location and depending on the application, if noisy data is a positive or negative. 9. If you wanted to evaluate the sensitivity of a sensor, you might choose a stable/static location and still hope for noisy data. If you wanted to evaluate the filtering performance of a sensor, you might choose a noisy/unpredictable environment and hope to see attenuation/very little noise in your data. For either scenario, you should still isolate the sensor from multiple sources of noise in order to verifiably interpret its performance for a single type.

Table of B, K, N Parameters for Allan Variance of Location D

	Gyro X	Gyro Y	Gyro Z	Accel X	Accel Y	Accel Z
B	1.109e-05	2.633e-05	5.281e-06	3.035e-04	0.0005155	6.719e-04
N	9.681e-05	1.220e-04	9.823e-05	0.0020	0.001843	0.00269
K	2.357e-07	1.988e-06	2.357e-07	2.372e-05	6.701e-05	7.811e-05

Appendix A – Collected Data

Figure 1: Stationary Roll, Pitch, Yaw Distributions

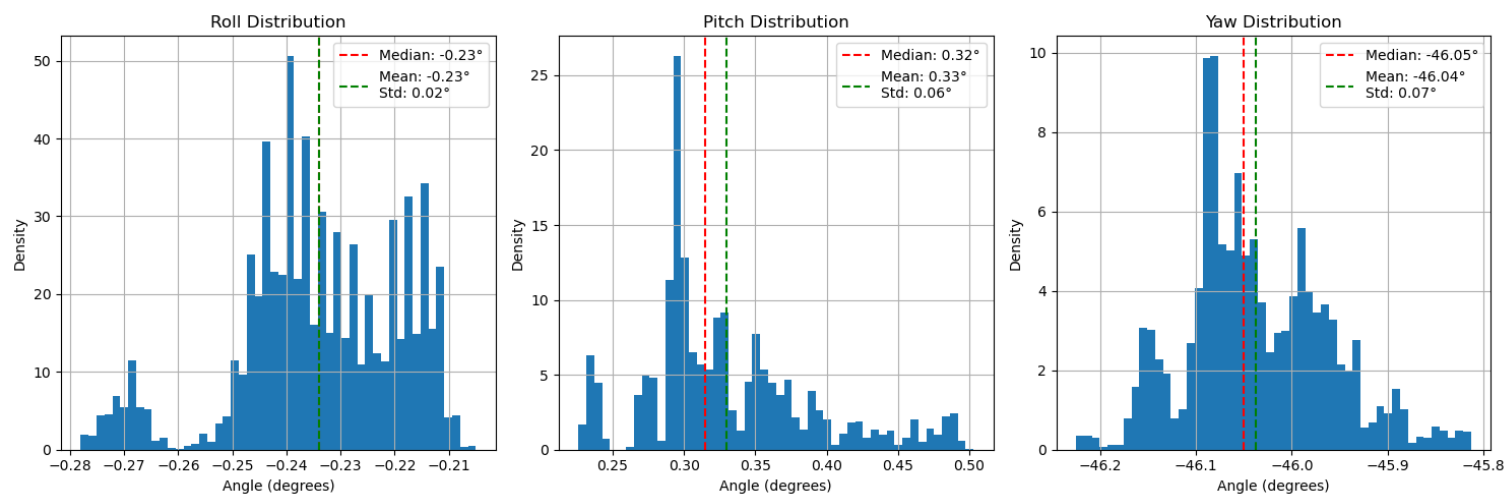
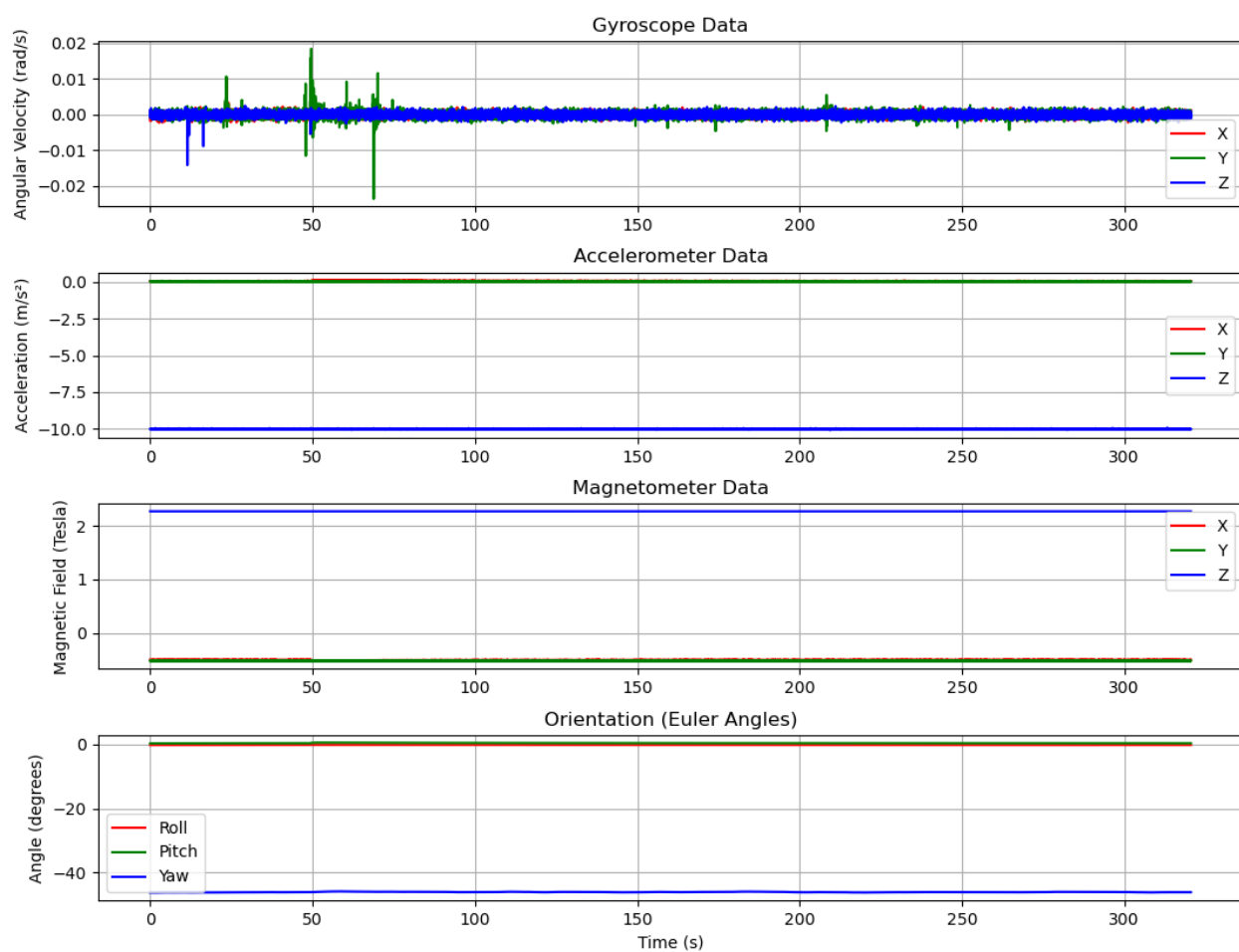


Figure 2: Stationary Gyroscope, Accelerometer, Magnetometer, & Euler Angles Over Time



Appendix B – Collected Data

Figure 3: Circular Motion about Y-Axis, Accelerometer Data

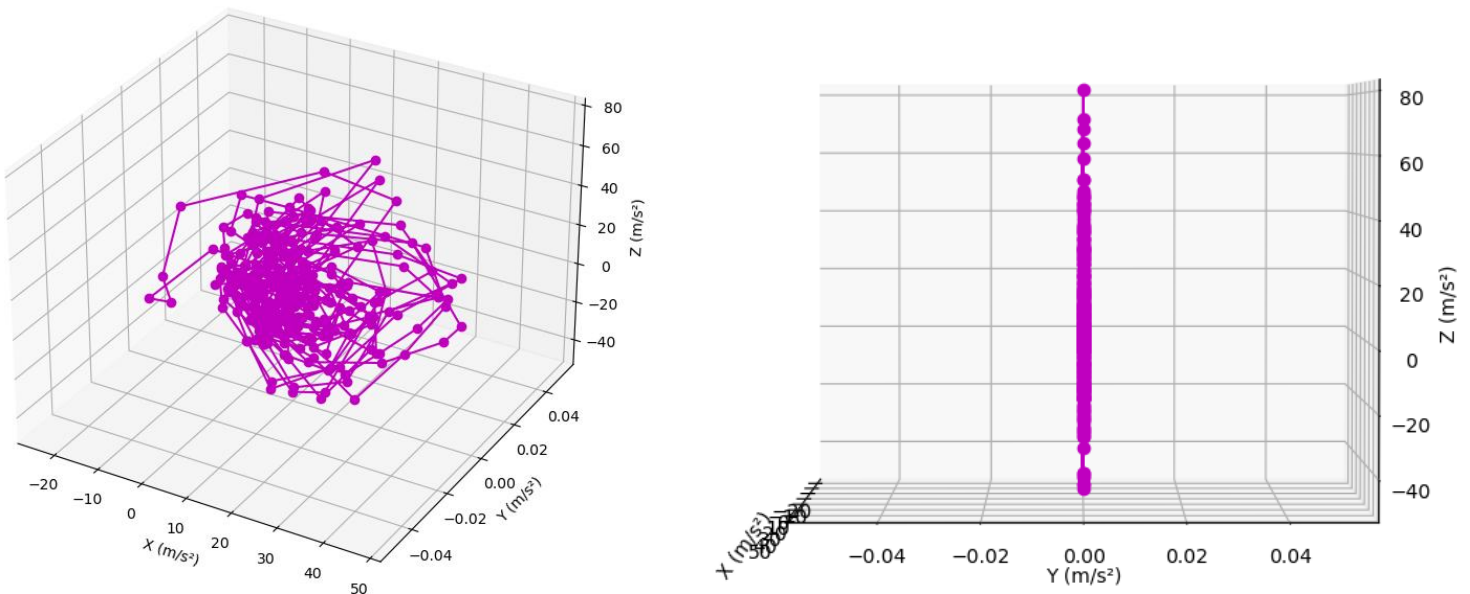
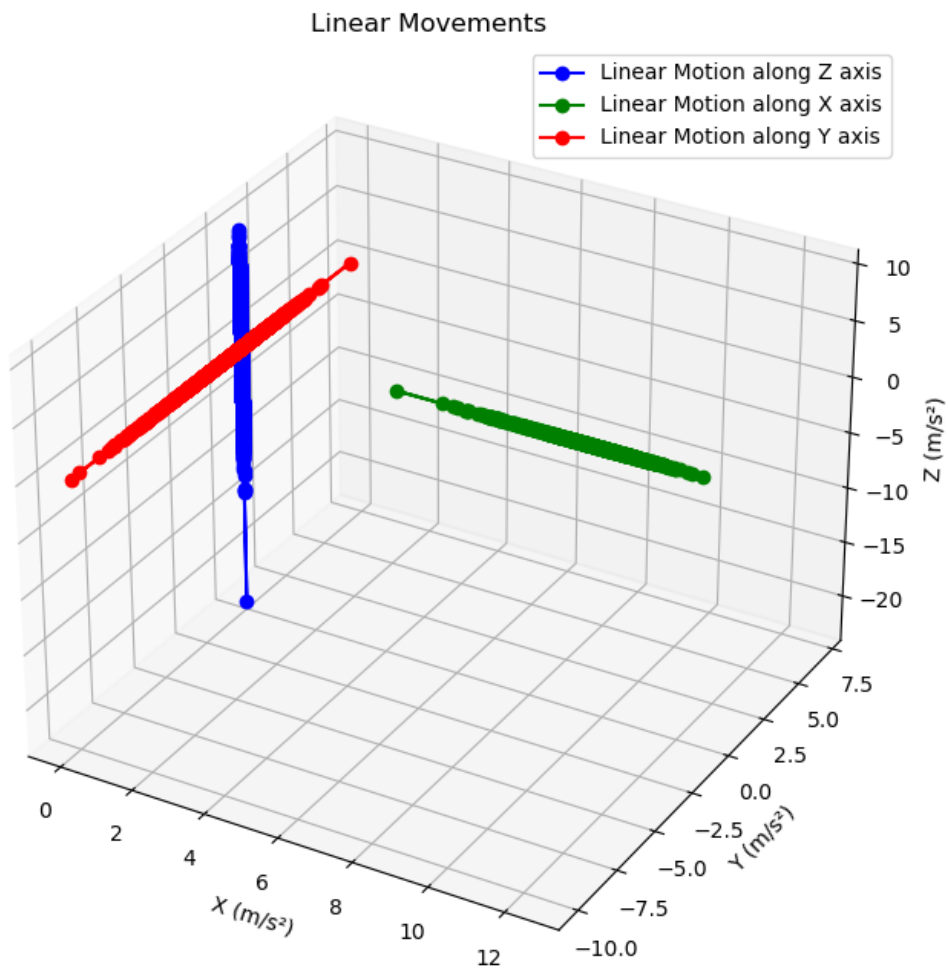
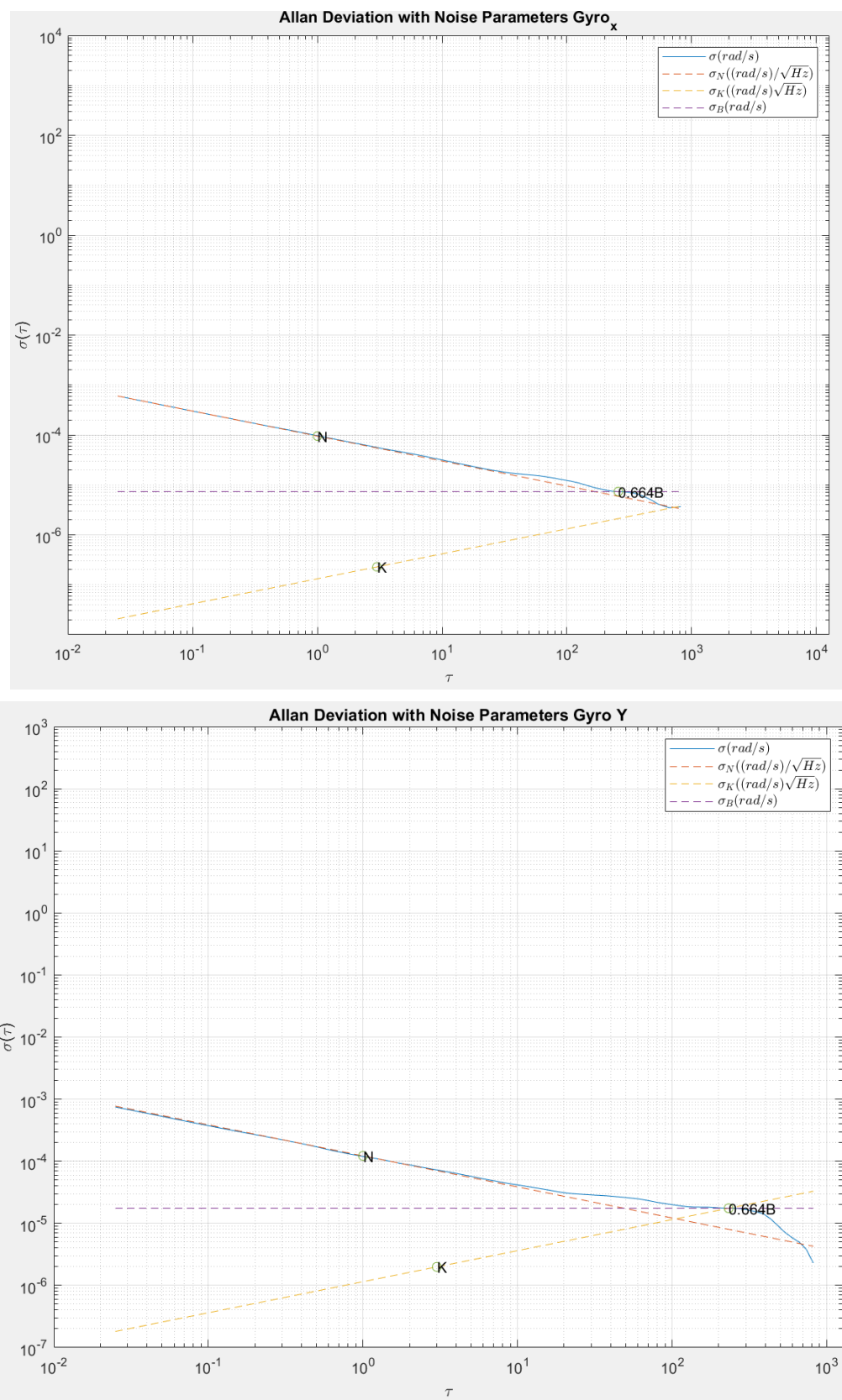


Figure 4: Linear Motion In X, Y, & Z Axes, Accelerometer Data



Appendix C – Large Stationary Datasets

Figure 5: Location D, Allan Variances of Gyroscope Data in X, Y, & Z Axes



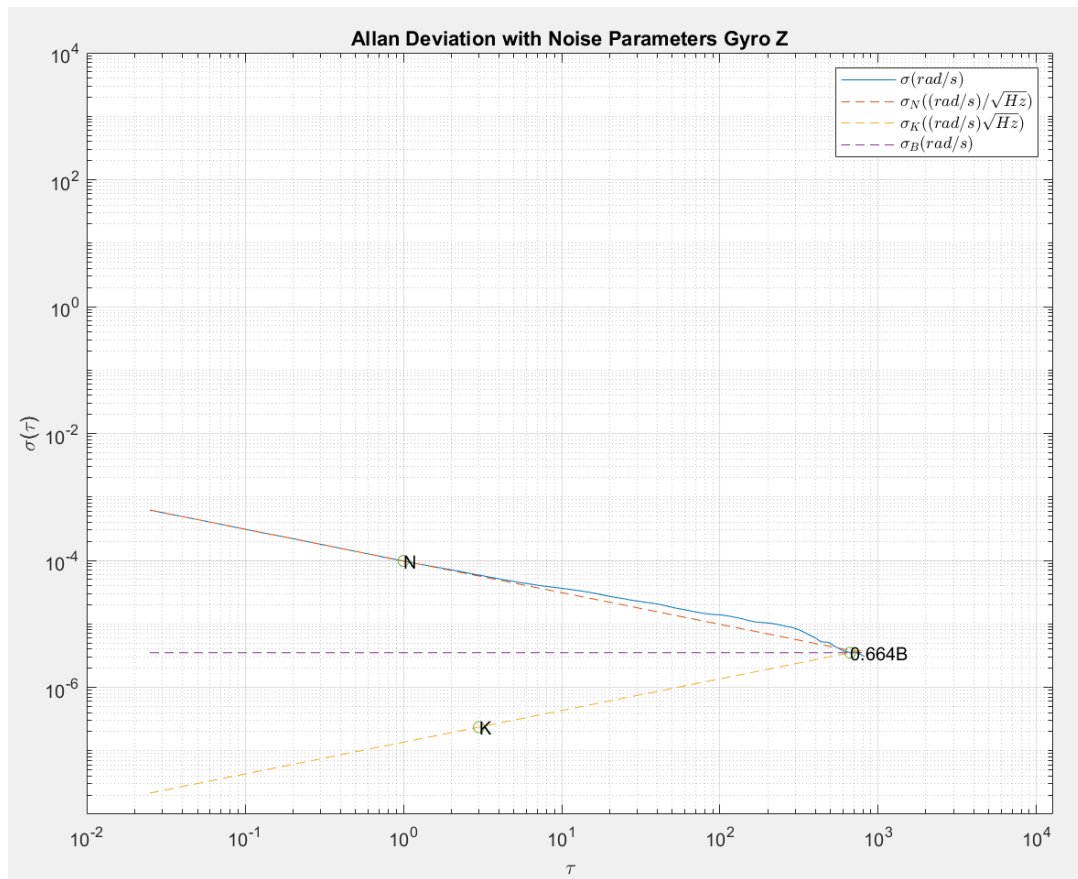
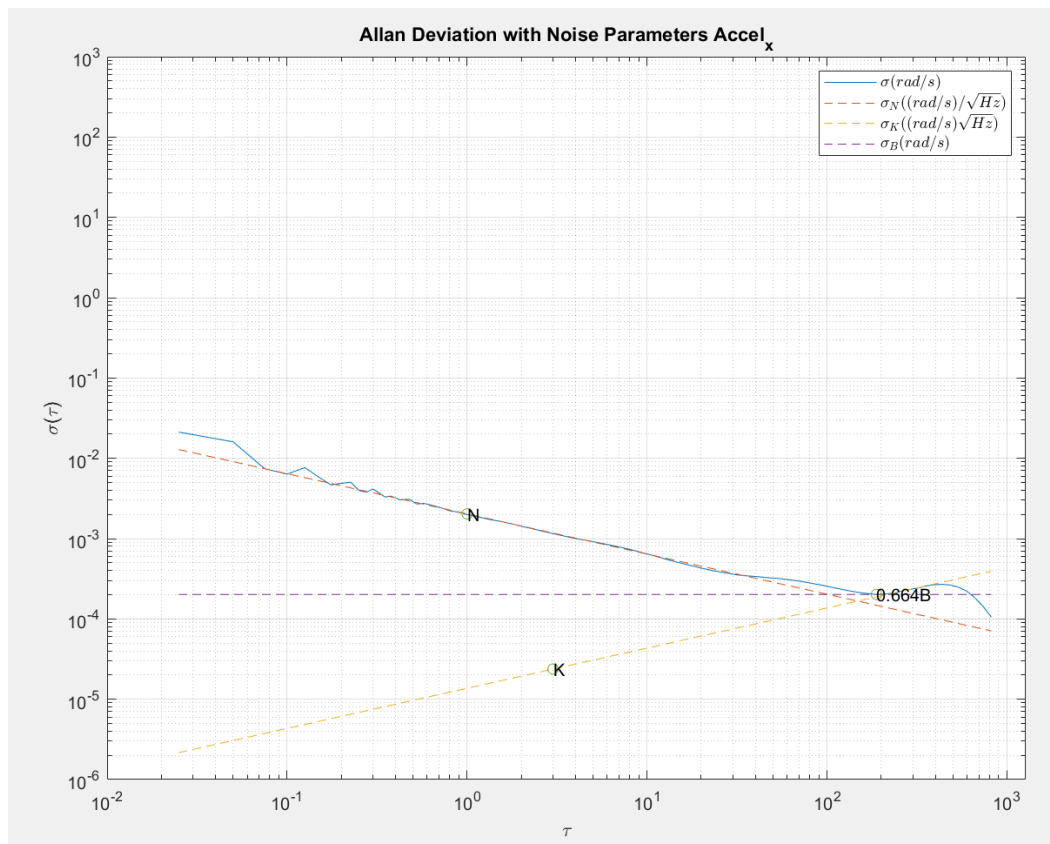
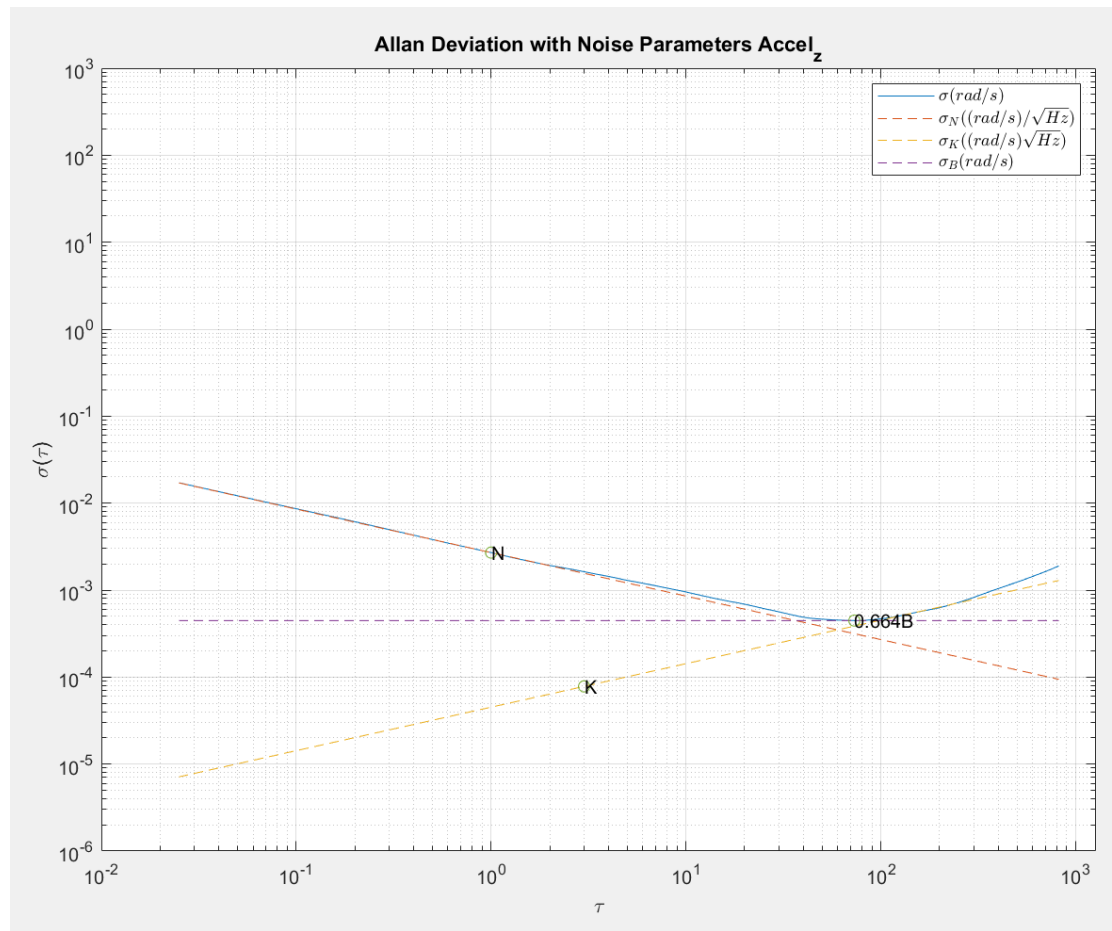
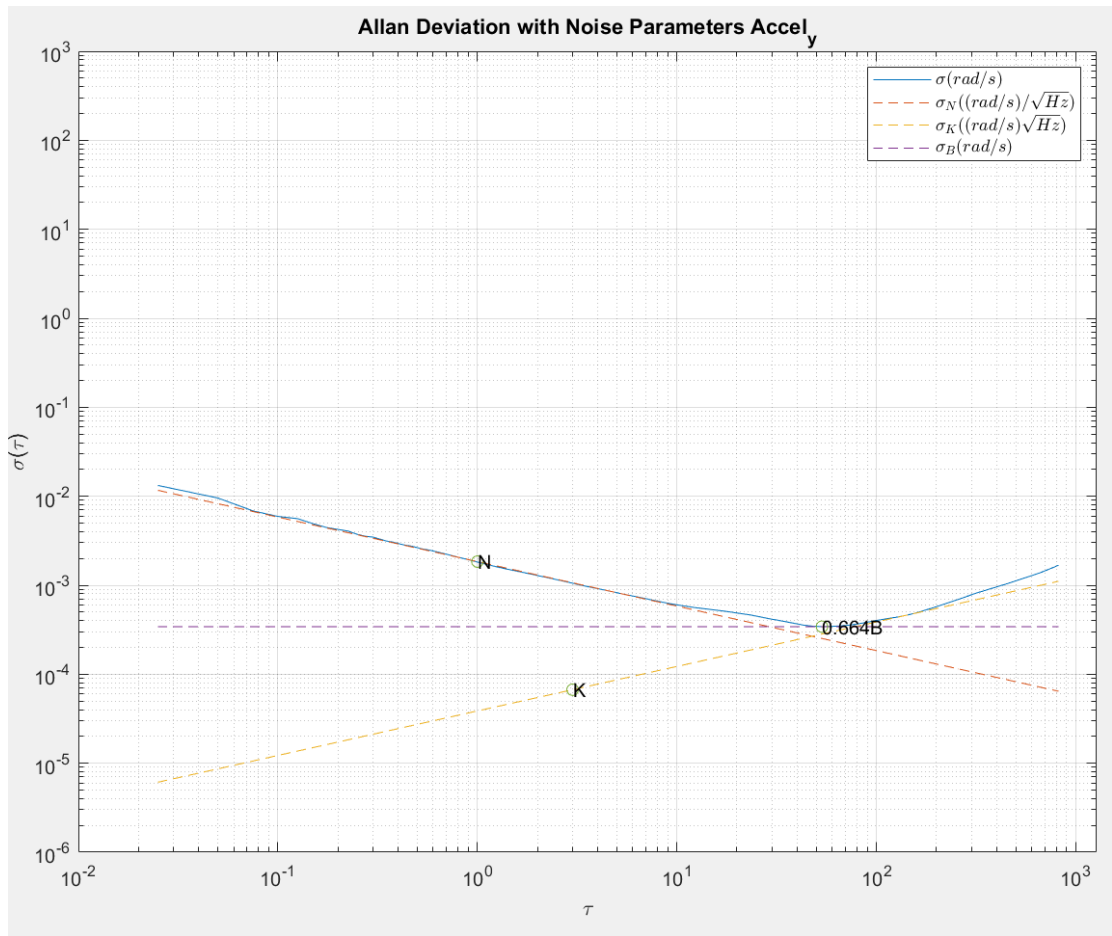
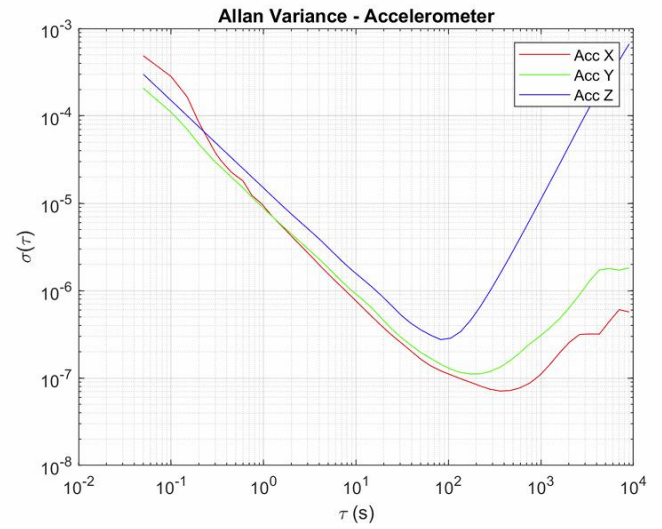
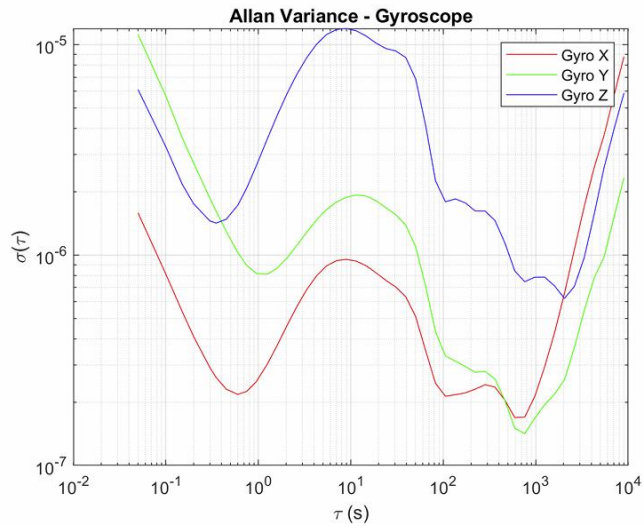


Figure 6: Location D, Allan Variances of Accelerometer Data in X, Y & Z Axes

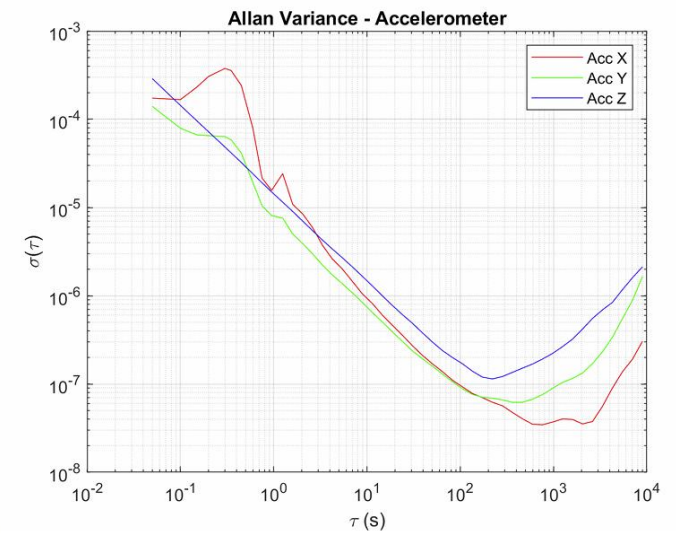
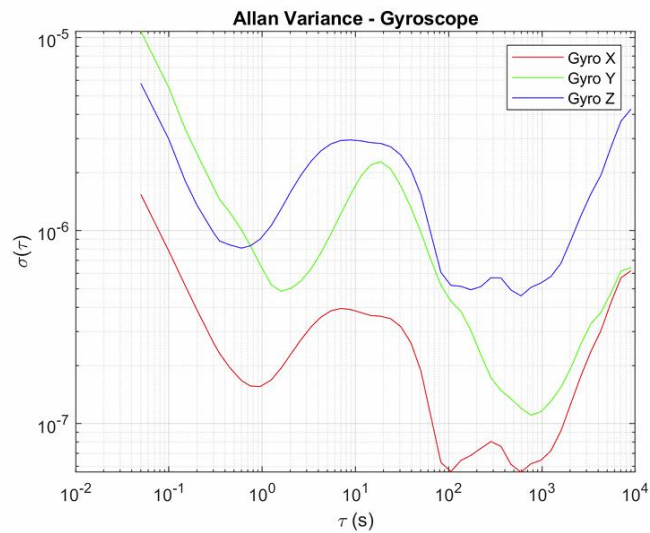




Location A



Location B



Location C

